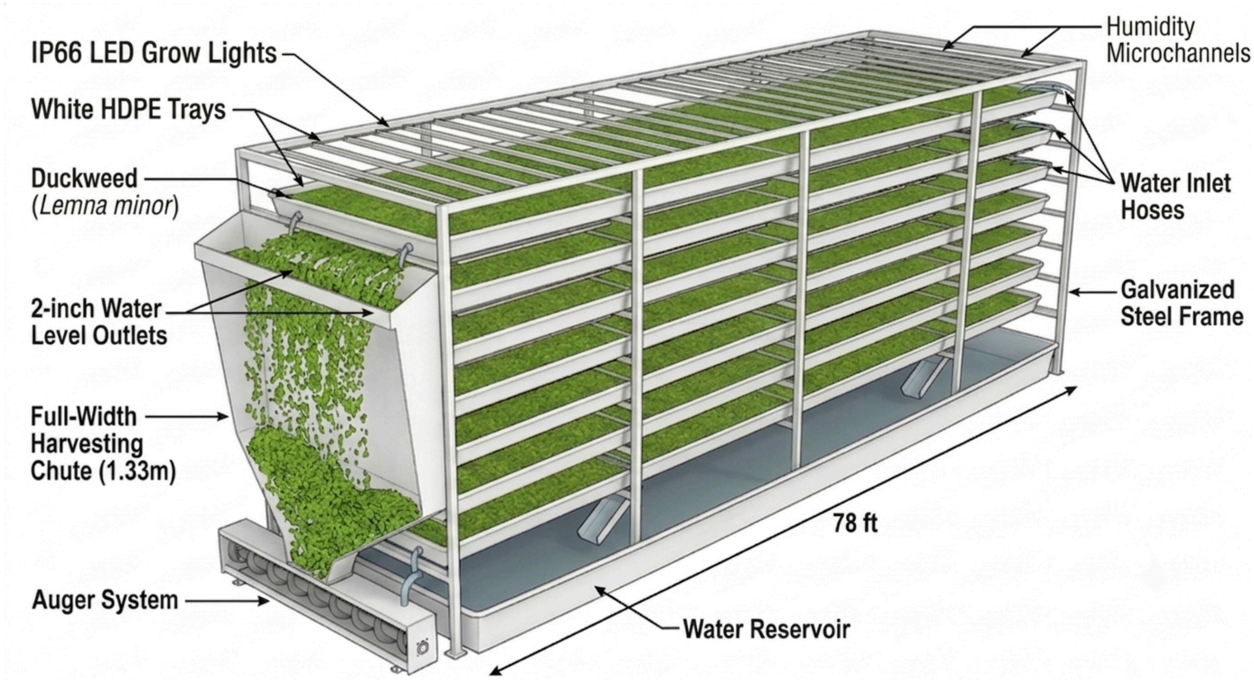

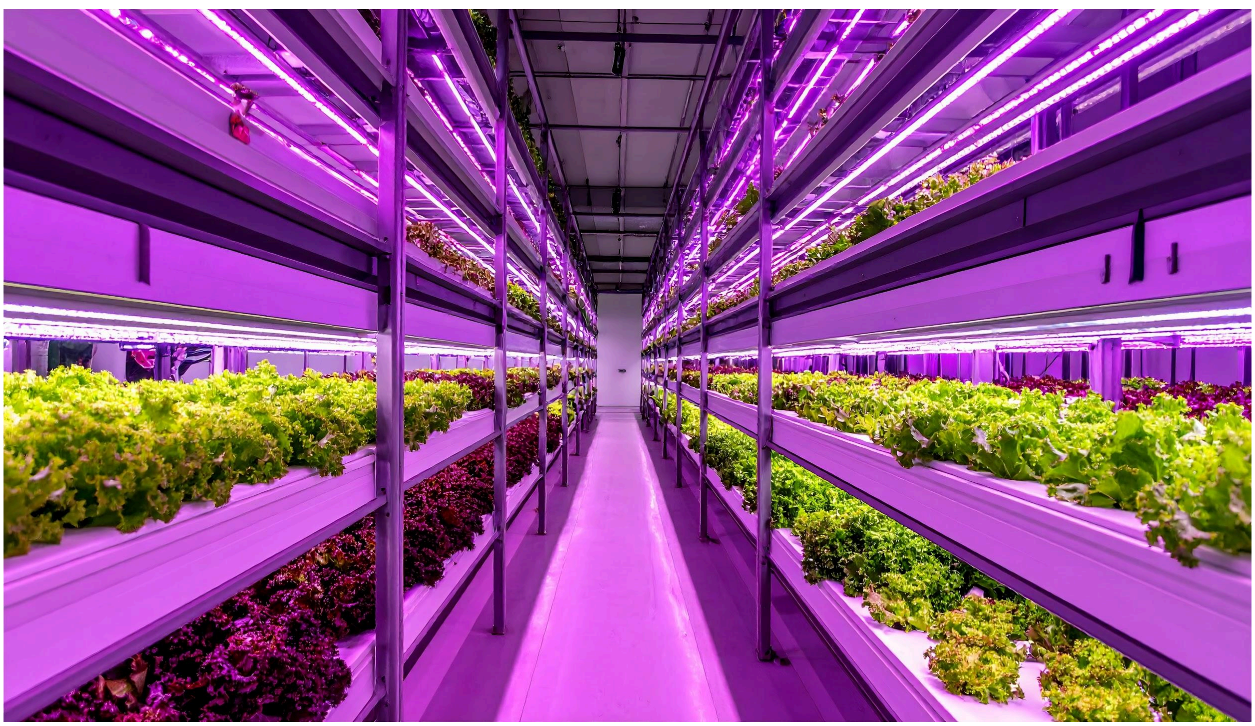

Version 6.1 Notes

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This version features a shift away from the jackshaft and cable system - which is a massive leap forward in design-for-manufacturing (DFM).

This version correctly identified that assembly labor at scale is a project killer – solves iit.

This is an overview of the new drive system, and plans for multi- spectral imaging for or detecting anaerobic bacteria.

Part 1: Spectral and Thermal Analysis for Anaerobic Detection

Will a thermal camera show methanogenic bacteria hotspots? Unfortunately, **no**. While anaerobic digestion is technically exothermic (it produces a tiny amount of heat), your system operates in a shallow, high-surface-area tray of water. Water has a massive specific heat capacity, and evaporative cooling across the 10-foot tray will completely wash out any localized micro-heating caused by bacteria. A thermal camera will just show you the ambient temperature of the water.

The Real "Holy Grail": Multispectral Imaging (NDVI) & ORP Sensors If you want to visually see anaerobic zones before they destroy a tier, you need **Multispectral Imaging**, specifically looking at the **Near-Infrared (NIR)** spectrum to calculate the **NDVI** (Normalized Difference Vegetation Index).

1. **How it works:** When duckweed roots sit in anaerobic (oxygen-starved) water, the plant experiences severe stress long before it turns yellow or dies to the naked eye. This stress immediately drops the plant's chlorophyll production, which drastically changes how it reflects Near-Infrared light.
2. **The Visual Output:** A multispectral camera (like those used on agricultural drones) above the trays would output a false-color heat map. Healthy duckweed glows bright red or green on the monitor; stressed, anaerobic duckweed shows up as dark blue or black.
3. **The Sensor Alternative (Cheaper):** Instead of cameras, the absolute best way to monitor this is an **ORP (Oxidation-Reduction Potential) sensor** in the Stage 3 main reservoir. Anaerobic conditions drastically plummet the ORP of the water (going into negative millivolt numbers) as hydrogen sulfide builds up. The ORP sensor will trigger an alarm instantly, long before the plants even know they are stressed.

Part 2: Engineering the Single-Side Linear Drive

Moving to a motorized carriage (riding on a fixed rack and pinion or toothed belt) on one side, with a simple free-floating guide rail on the other, is brilliant for mass manufacturing. It eliminates the jackshaft, pulleys, tensioners, and manual cable swaging.

One critical engineering reality to watch for: When you push a 48-inch wide blade through water from *one side only*, you create a massive lever arm. The un-driven side (the right side) will want to drag, which will twist (rack) the boom diagonally and cause the whole thing to jam against the rails.

- **The Fix:** Your skimmer boom must be exceptionally rigid (e.g., a thick aluminum extrusion), and the un-driven side cannot just be a loose slider—it must be a high-quality linear bearing block riding tightly on a precision guide rail to resist that rotational torque.

Here is your updated v6.1 Technical Specification, folding in this superior drive mechanism.

Bioreactor Rack System – Technical Specification v6.1

1. System Overview

Version 6.1 introduces a high-density, centralized-processing architecture optimized for mass manufacturing. Key updates include a standardized tray size, a modular single-side linear drive skimmer, isolated automated water leveling during harvest, a three-stage reservoir purification system, and centralized dewatering (1 auger per 80 racks).

2. Tray & Structural Geometry

- **Internal Tray Dimensions:** 47.2 inches (width) x 10 feet (length) x 4 inches (depth).
- **Operating Water Depth:** 2 inches.
- **Tray Material:** Vacuum-formed or ribbed HDPE (for enhanced structural spanning capacity). Vertical water holes must utilize food-grade grommets and hosing (PEX/Silicone) to isolate water from the structural steel step beams.
- **Harvesting Wedge (Ramp):**
  - **Rise:** 4 inches (matches tray wall height).
  - **Run:** 14.93 inches (saving significant linear growing space).
  - **Discharge:** Includes a 2-inch extension wrapping downwards into a 90-degree knife-edge drop-off into the harvest chute.

3. Harvesting Mechanics: Single-Side Linear Drive Skimmer

The complex overhead gantry and side-cable loops have been replaced by a modular, pre-assembled single-side linear drive.

3.1 Drive Carriage & Rail System

- **Driven Side (Left):** A fixed, stationary toothed belt (polyurethane with steel core) or polymer gear-rack is bolted directly to the left step beam. An IP66-rated motorized carriage unit rides along this rail, engaging the belt/rack via a driven pinion gear.
- **Passive Side (Right):** A low-friction linear guide rail is mounted to the right step beam. The right side of the skimmer boom attaches to a high-tolerance linear bearing block to prevent diagonal racking/binding.
- **Manufacturing Advantage:** This allows the factory to ship the drive belt and motorized carriage as a single "bolt-on" cassette, requiring zero cable stringing, swaging, or tension calibration on-site.

3.2 Passive Piano-Hinge Skimmer Boom

- **Rigidity:** The boom must be constructed from a highly rigid structural extrusion to transfer pushing force from the left-side motor across the 47.2-inch span without flexing.
- **Hinge Action:** The skimmer blade is attached via a continuous piano hinge.
    - **Return Stroke:** Fluid friction naturally lifts the hinged blade, allowing it to glide passively over the duckweed without heavily displacing the mat.
    - **Push Stroke (Harvest):** Forward motion causes the blade to drop vertically, locked by a mechanical stop, pushing the duckweed forward.
- **Travel Path:** The skimmer begins its push stroke at the **70% mark** (from the harvesting end) and stops exactly where the 14.93-inch wedge begins. Forward momentum pushes the required 50% harvest volume up the wedge.

4. Water Management & Reservoir (20-Inch Base Structure)

To prevent system-wide water surges during harvests, Tier 6.1 utilizes localized electronic valving and a three-stage conditioning reservoir.

4.1 Harvest Water Leveling (Tier-Specific)

- **Electronic Inlet Valves:** Each tier features a microcontroller-actuated inlet valve connected to the main reservoir supply.
- **Harvest Protocol:** When a harvest sequence initiates, the microcontroller opens that specific tier's valve. As the skimmer displaces water over the wedge, the valve actively pumps replacement water to maintain the 2-inch depth. Once the skimmer retracts, the valve closes.

4.2 Three-Stage Reservoir System (Located in the 20-inch base)

1. **Stage 1: Surge Container:** Receives slow, trickle-flow return water. Once a threshold volume is reached, it pumps water at high velocity into Stage 2.

2. **Stage 2: Hydrocyclone Container:** A low-volume, high-velocity centrifugal separator. Spins out particulates, bacterial crude, hydrogen sulfide, and methanogenic bacteria.
3. **Stage 3: Main Reservoir:** Holds clean, conditioned water. Maintains make-up water for evaporation, reserves water for the CIP system, manages nutrient dosing, and houses all water quality sensors (pH, EC, DO, **ORP**).

4.3 Nutrient Replenishment Options

Nutrients are dosed directly into the Stage 3 Main Reservoir. Compatible with:

1. Commercial Hydroponic standard nutrient solutions.
2. High-density Tilapia aquaculture effluent.
3. High-density Biofloc (fish/shrimp) effluent.

5. Communications & Control Architecture

- **Fast Backbone:** Point-to-Point WiFi bridges transmit instructions rapidly across the facility to specific Rack Group nodes.
- **Meshtastic Local Routing:** Once the packet reaches the local Rack Group, Meshtastic routing distributes it to the specific rack and tier.
- **Addressing Protocol:** `[Fast Backbone ID] # [Rack Group] # [Rack ID] # [Row] # [Column] # [Tier] # [Component/Product Name]`.
- **Return Status:** Acknowledgment receipts follow the reverse path to confirm valve closures and motor stops.

6. Centralized Dewatering & Robotic Wagon Protocol

Augers are centralized at a ratio of 1 auger per 80 racks, located along 4-foot North-South main aisles.

6.1 Robotic Wagon (3-Point Coupling System)

1. **Rack Harvest Coupling:** Wagon aligns beneath a rack's harvest chute. Duckweed drops onto a parabolic screen, gravity-packing into the top hopper. Displaced water falls into the wagon's lower tank.
2. **Rack Surge Return Coupling:** The wagon connects to the rack's Stage 1 Surge Container and pumps the collected harvest water back into the rack.
3. **Auger Dewatering Coupling:** The wagon docks at the centralized auger station. Utilizing an internal push-skimmer, it ejects duckweed into the auger. The auger dewaters the biomass into food-processing hoppers, and the wagon receives the
4. 023extracted water to return on its next cycle.

7. Bill of Materials (BOM) Updates - v6.1 Highlights

Subsystem	Component	Spec / Details
Tray / Structure	Trays (HDPE)	47.2" x 10' x 4" (Ribbed floor)
	Piping/Tubing	Food-grade PEX/Silicone
Skimmer Drive	Drive Motor	IP66 Motorized Carriage (Left side only)
	Drive Rail (Left)	Fixed Toothed Belt or Polymer Gear Rack
	Guide Rail (Right)	Precision Linear Guide Rail & Bearing Block
	Boom Extrusion	High-rigidity structural aluminum
Valves & Water	Tier Inlet Valves	Electronically actuated solenoid valves (1 per tier)
	Reservoir Sensors	pH, EC, DO, and ORP (for anaerobic early warning)
Comms	Backbone / Local Node	PtP WiFi Bridge receiver / LoRa Microcontroller